frontiers in Psychiatry

Classifying Autism Spectrum Disorder Using the Temporal Statistics of Resting-State Functional MRI Data With 3D Convolutional Neural Networks

Rajat Mani Thomas[*†], *Selene Gallo*[†], *Leonardo Cerliani*, *Paul Zhutovsky*, *Ahmed El-Gazzar* and *Guido van Wingen*

Department of Psychiatry, Amsterdam University Medical Center, Amsterdam, Netherlands

Mina Kwon

Clinical-AI Journal Club

2021.02.22

Computational Clinical Science Laboratory

Introduction

Autism Spectrum Disorder (ASD)

- Difficulty with communication and interaction with other people
- Restricted interests and repetitive behaviors
- Symptoms that affect the person's ability to function in school, work, and other areas of life

More details of ASD: https://www.nimh.nih.gov/health/publications/autism-spectrum-disorder/19-mh-8084-autismspecdisordr_152236.pdf
Image from: https://scitechdaily.com/scientists-reverse-autism-symptoms-in-mice/

Background knowledge

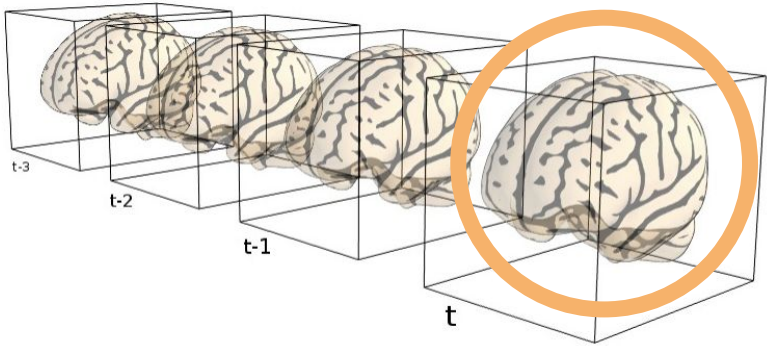
- Brain abnormalities of psychiatric disorders are reflected in **functional scans** rather than structural images (Woodward & Cascio, 2015).

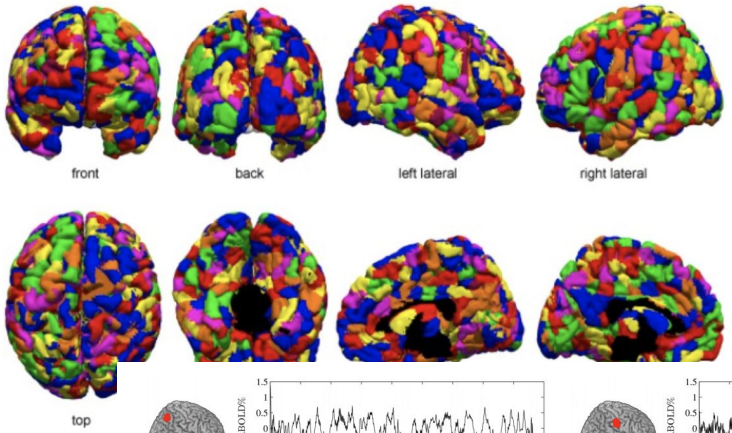
- Machine learning could not be fully utilized due to 1) lack of large datasets, and 2) high dimensionality of neuroimaging data.

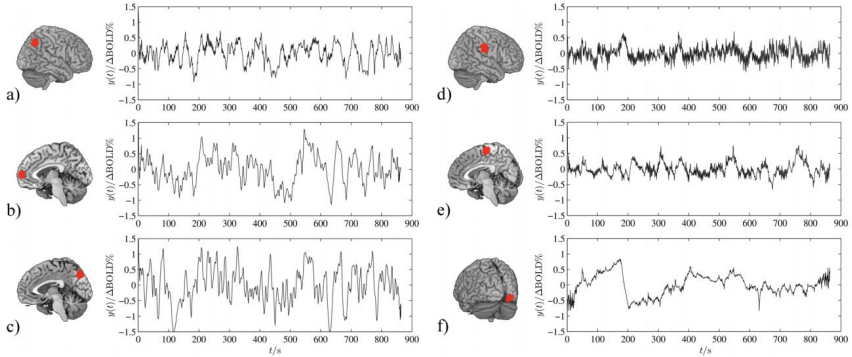
  1) Data aggregation
  2) Dimensionality reduction

Dimensionality Reduction

1) Spatial downscaling



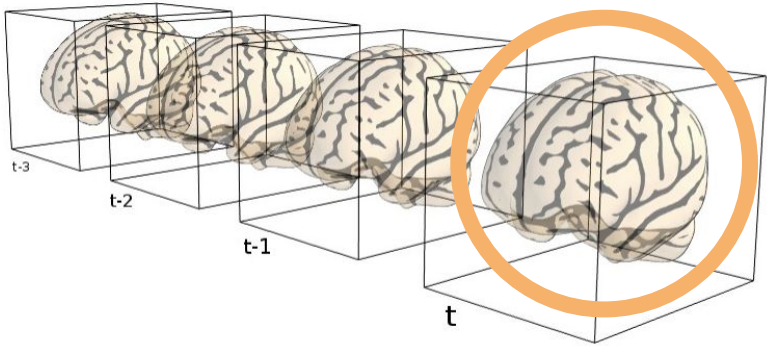




left: https://nilearn.github.io/building_blocks/manual_pipeline.html/
Right top: https://www.sciencedirect.com/science/article/pii/S016502701730376X/
Right bottom:
https://www.researchgate.net/publication/273664506_Synchronization_non-linear_dynamics_and_low-frequency_fluctuations_Analogy_between_spontaneous_brain_activity_and_networked_single-transistor_chaotic_oscillators

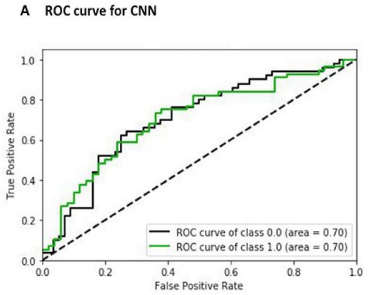
Dimensionality Reduction

1) Spatial downscaling



Automated Detection of Autism Spectrum Disorder Using a Convolutional Neural Network

Zeinab Sherkatghanad[1†], Mohammadsadegh Akhondzadeh[2†], Soorena Salari[3†], Mariam Zomorodi-Moghadam[4†], Moloud Abdar[5†], U. Rajendra Acharya[6,7,8], Reza Khosrowabadi[9] and Vahid Salari[1,10]*



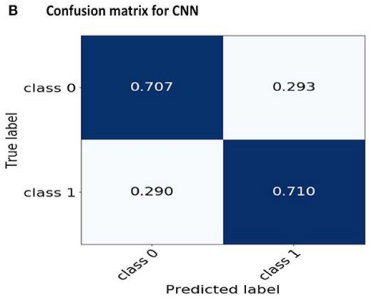
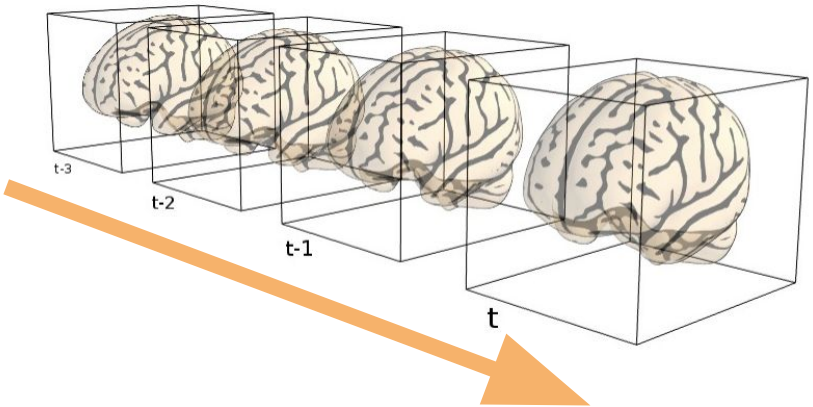
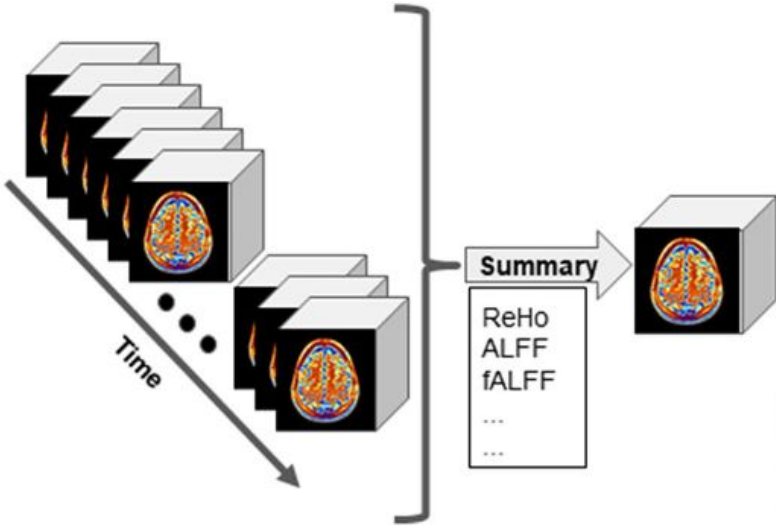


Image reference: https://nilearn.github.io/building_blocks/manual_pipeline.html

Dimensionality Reduction



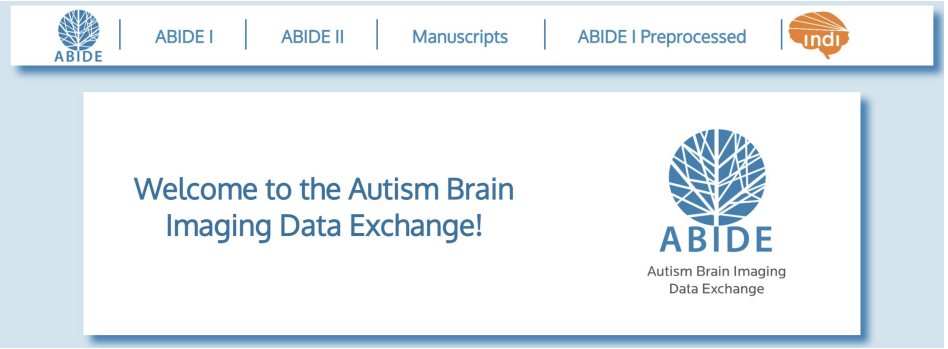


2) summarizing time domain

Left: https://nilearn.github.io/building_blocks/manual_pipeline.html / Right: Figure 1 of this article

Methods

Autism Brain Imaging Data Exchange Dataset



⇒ N = 2,169

1,028 Autism Spectrum Disorders (ASD)

1,141 controls (CON)

After quality check & subject selection,
⇒ 1, 162 subjects (620 ASD, 542 CON)

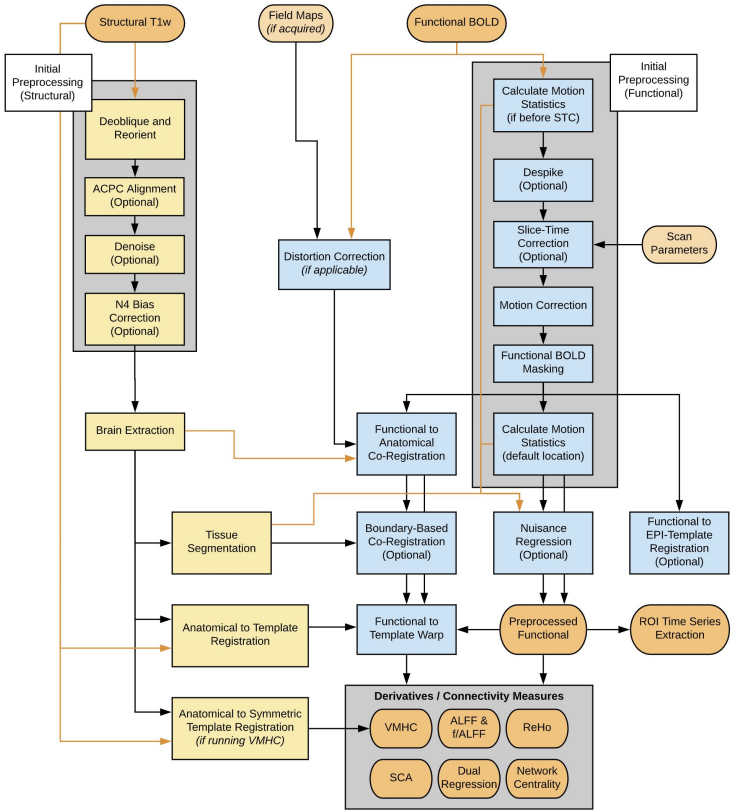
ABIDE: http://fcon_1000.projects.nitrc.org/indi/abide/

Preprocessing

Configurable Pipeline for the Analysis of Connectomes (C-PAC):
https://fcp-indi.github.io/

GitHub repo:
https://github.com/FCP-INDI/C-PAC

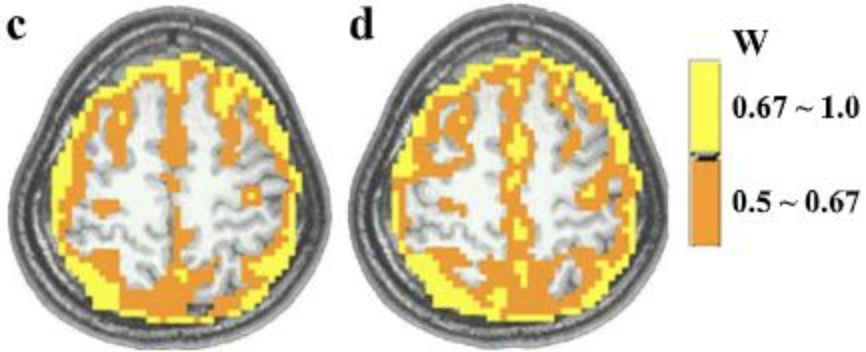
Preprocessing steps:
https://fcp-indi.github.io/docs/latest/user/preprocessing



9 Summary measures of rs-fMRI

1. Regional homogeneity (ReHO)

: similarity between time series of a given voxel and its neighbor, calculated by Kendall's coefficient of concordance (KCC)



Zang et al. (2004) https://doi.org/10.1016/j.neuroimage.2003.12.030

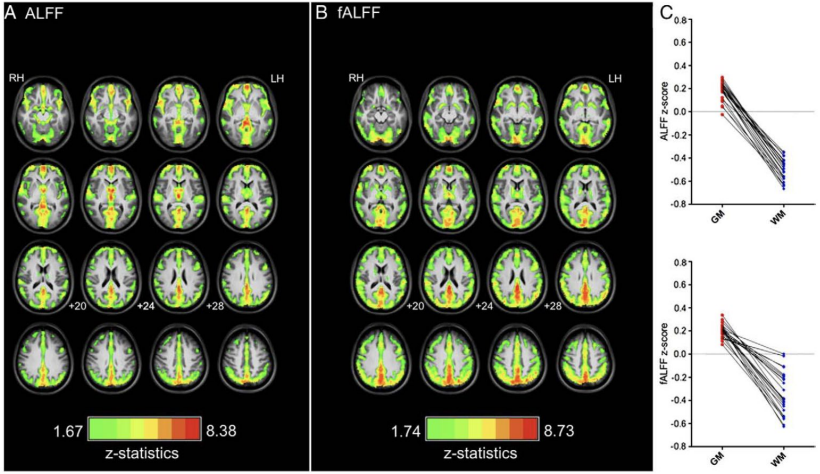
9 Summary measures of rs-fMRI

2. Amplitude of low frequency fluctuations (ALFF)

: Indexes strength of low frequency oscillations, detecting regional intensity of spontaneous fluctuation of BOLD signal

3. Fractional ALFF (fALFF)

: Normalized index of ALFF



Zuo et al. (2010) https://doi.org/10.1016/j.neuroimage.2009.09.037

9 Summary measures of rs-fMRI

4. Degree centrality (DC)

: measures local network connectivity by counting edges of each node

5. Eigenvector centrality (EC)

: measures global network connectivity by counting edges with node of high centrality

6. Local functional connectivity density (LFCD)

: pearson linear correlation coefficient between voxels

DC & EC: Zuo et al. (2012) https://doi.org/10.1093/cercor/bhr269
LFCD: Tomasi & Volkow (2011) https://doi.org/10.1093/cercor/bhq268

9 Summary measures of rs-fMRI

7. Entropy

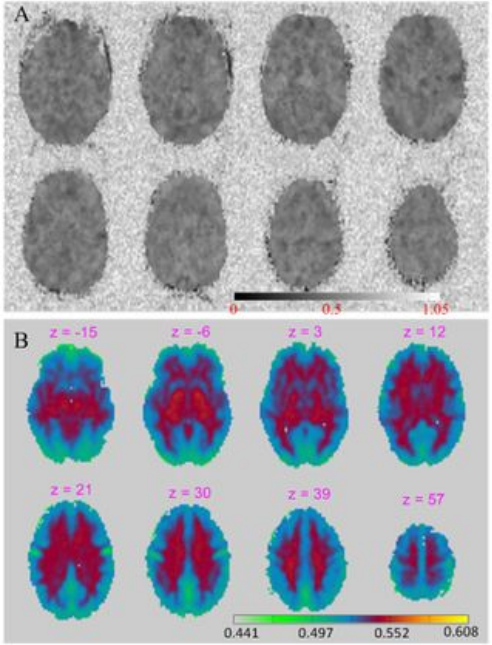
: measures predictability/irregularity of brain,

8. Voxel-mirrored homotopic connectivity (VMHC)

: voxel-wise measure of connectivity between hemispheres

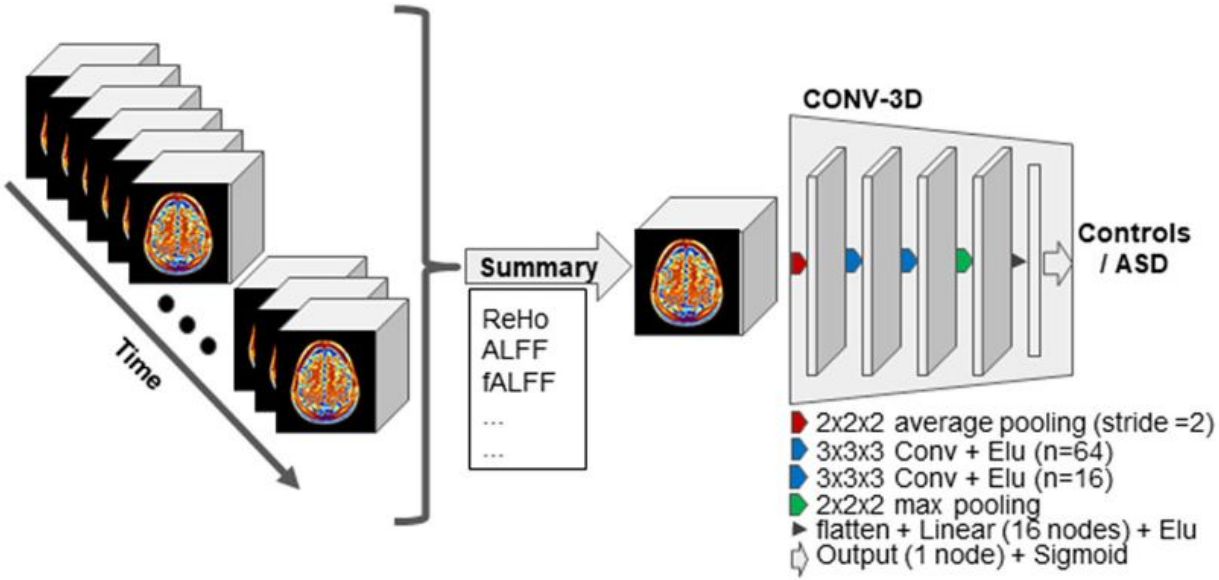
9. Auto-correlation log

: correlation between the voxel's past and present states.



Entropy: Wang et el. (2014) https://doi.org/10.1371/journal.pone.0089948
Auto-correlation log: Opelt et al. (2004) https://doi.org/10.1371/journal.pone.0038131

3D Convolutional Neural Network (3D CNN)



For more information about this model: Khosla et al. (2019) https://doi.org/10.1016/j.neuroimage.2019.06.012
For basic understanding of CNN: https://www.youtube.com/watch?v=YRhxdVk_sIs

Models

1. Single-measure Model (SM-model)
: 9 models with each summary measure.

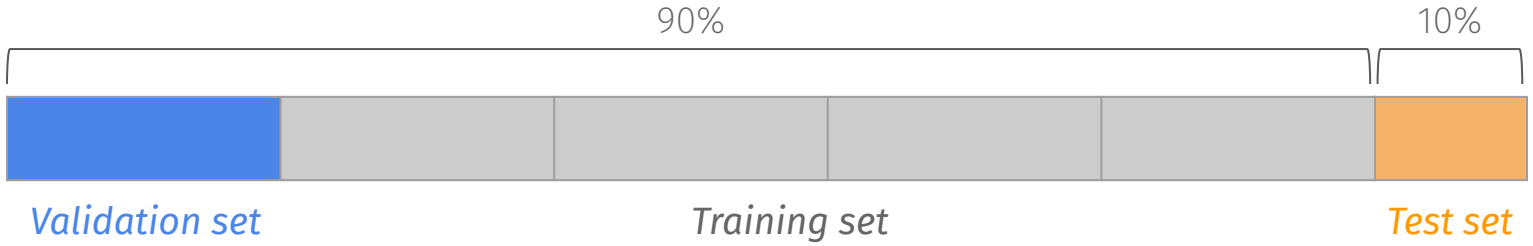
2. Multi-measure Ensemble (MM-ensemble)
: Train 9 models with each summary measure which *vote* for final classification.
⇒ Majority vote used as predicted outcome.

3. Multi-measure Model (MM-model)
: Single model with 9 summary measures combined as input

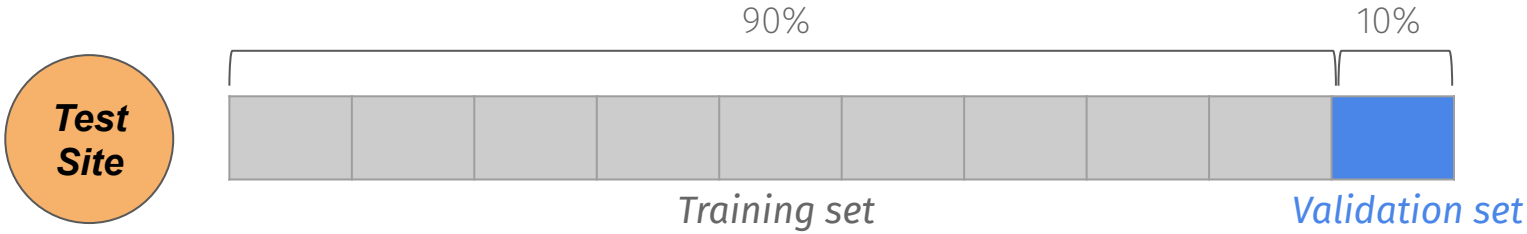
⇒ For CNN & Linear Support vector machine algorithm (SVM)

Cross-validation (CV) procedure

1. 5-fold CV



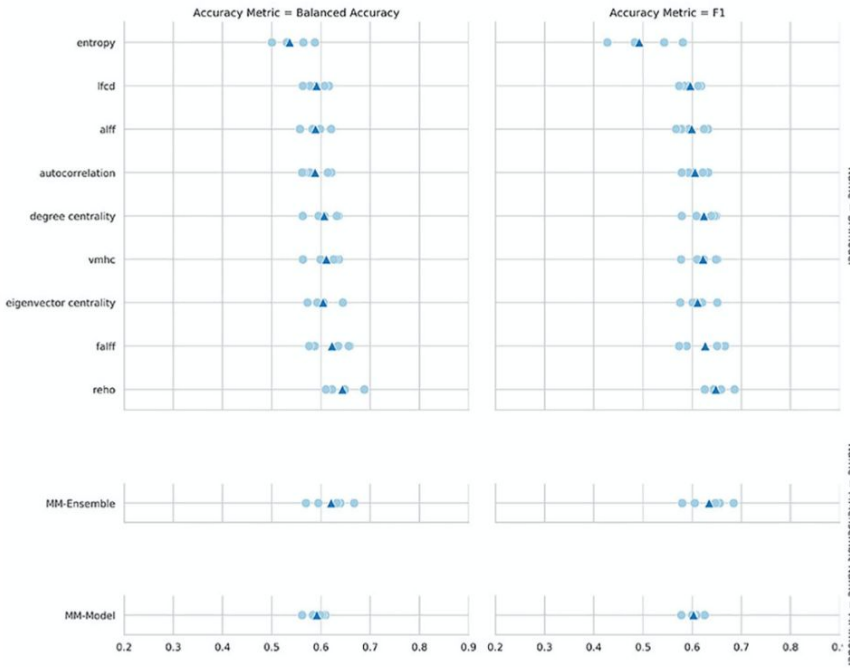
2. Leave-site-out CV



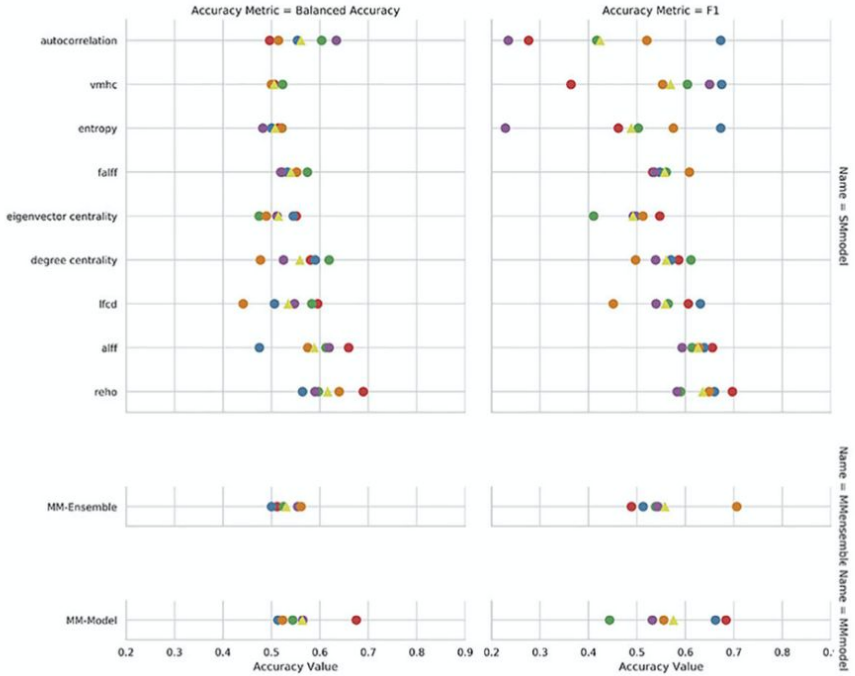
Five test sites with highest number of subjects

Result

3D-CNN Results



5-fold CV



Leave-site-out CV

3D-CNN Results

***MM ensemble** showed highest prediction accuracy in 5-fold CV, but not in Leave-site-out CV*

	3D-CNN		SVM	
	Accuracy	**F1-score**	**Accuracy**	**F1-score**
ReHo	**0.64**	0.65	0.66	0.66
fALFF	0.62	0.63	0.57	0.58
Degree centrality	0.61	0.62	0.63	0.64
VMHC	0.61	0.62	0.62	0.62
Eigenvector centrality	0.60	0.61	0.61	0.63
Autocorr lag	0.59	0.61	0.57	0.59
LFCD	0.59	0.60	0.63	0.65
ALFF	0.59	0.60	0.57	0.58
Entropy	0.54	0.49	0.56	0.57
MM ensemble	**0.64**	**0.66**	**0.66**	**0.67**
MM model	0.59	0.60	0.61	0.62

In bold = Highest score in that column.

5-fold CV

	3D-CNN		SVM	
	Accuracy	**F1-score**	**Accuracy**	**F1-score**
ReHo	**0.62**	**0.64**	**0.63**	**0.62**
fALFF	0.59	0.63	0.61	0.61
VMHC	0.56	0.57	0.62	0.59
Degree centrality	0.56	0.56	0.60	0.59
fALFF	0.54	0.56	0.57	0.54
LFCD	0.53	0.56	0.63	0.61
Eigenvector centrality	0.51	0.49	0.59	0.57
Entropy	0.51	0.49	0.53	0.52
Autocorr	0.51	0.42	0.54	0.55
MM ensemble	0.56	0.59	0.53	0.56
MM model	0.56	0.58	0.61	0.62

*See **Figure 3** and **Supplementary Tables S1** and **S2** for details on how each test site performed.*
In bold = Highest score in that column.

Leave-site-out CV

3D-CNN Results

***ReHo** showed highest performance in both CV procedures.*

	3D-CNN		SVM	
	Accuracy	**F1-score**	**Accuracy**	**F1-score**
ReHo	**0.64**	0.65	0.66	0.66
fALFF	0.62	0.63	0.57	0.58
Degree centrality	0.61	0.62	0.63	0.64
VMHC	0.61	0.62	0.62	0.62
Eigenvector centrality	0.60	0.61	0.61	0.63
Autocorr lag	0.59	0.61	0.57	0.59
LFCD	0.59	0.60	0.63	0.65
ALFF	0.59	0.60	0.57	0.58
Entropy	0.54	0.49	0.56	0.57
MM ensemble	**0.64**	**0.66**	**0.66**	**0.67**
MM model	0.59	0.60	0.61	0.62

In bold = Highest score in that column.

5-fold CV

	3D-CNN		SVM	
	Accuracy	**F1-score**	**Accuracy**	**F1-score**
ReHo	**0.62**	**0.64**	**0.63**	**0.62**
fALFF	0.59	0.63	0.61	0.61
VMHC	0.56	0.57	0.62	0.59
Degree centrality	0.56	0.56	0.60	0.59
fALFF	0.54	0.56	0.57	0.54
LFCD	0.53	0.56	0.63	0.61
Eigenvector centrality	0.51	0.49	0.59	0.57
Entropy	0.51	0.49	0.53	0.52
Autocorr	0.51	0.42	0.54	0.55
MM ensemble	0.56	0.59	0.53	0.56
MM model	0.56	0.58	0.61	0.62

Leave-site-out CV

3D-CNN Results

SVM confirmed the results of 3D-CNN

	3D-CNN		SVM	
	Accuracy	**F1-score**	**Accuracy**	**F1-score**
ReHo	**0.64**	0.65	0.66	0.66
fALFF	0.62	0.63	0.57	0.58
Degree centrality	0.61	0.62	0.63	0.64
VMHC	0.61	0.62	0.62	0.62
Eigenvector centrality	0.60	0.61	0.61	0.63
Autocorr lag	0.59	0.61	0.57	0.59
LFCD	0.59	0.60	0.63	0.65
ALFF	0.59	0.60	0.57	0.58
Entropy	0.54	0.49	0.56	0.57
MM ensemble	**0.64**	**0.66**	**0.66**	**0.67**
MM model	0.59	0.60	0.61	0.62

In bold = Highest score in that column.

5-fold CV

	3D-CNN		SVM	
	Accuracy	**F1-score**	**Accuracy**	**F1-score**
ReHo	**0.62**	**0.64**	**0.63**	**0.62**
fALFF	0.59	0.63	0.61	0.61
VMHC	0.56	0.57	0.62	0.59
Degree centrality	0.56	0.56	0.60	0.59
fALFF	0.54	0.56	0.57	0.54
LFCD	0.53	0.56	0.63	0.61
Eigenvector centrality	0.51	0.49	0.59	0.57
Entropy	0.51	0.49	0.53	0.52
Autocorr	0.51	0.42	0.54	0.55
MM ensemble	0.56	0.59	0.53	0.56
MM model	0.56	0.58	0.61	0.62

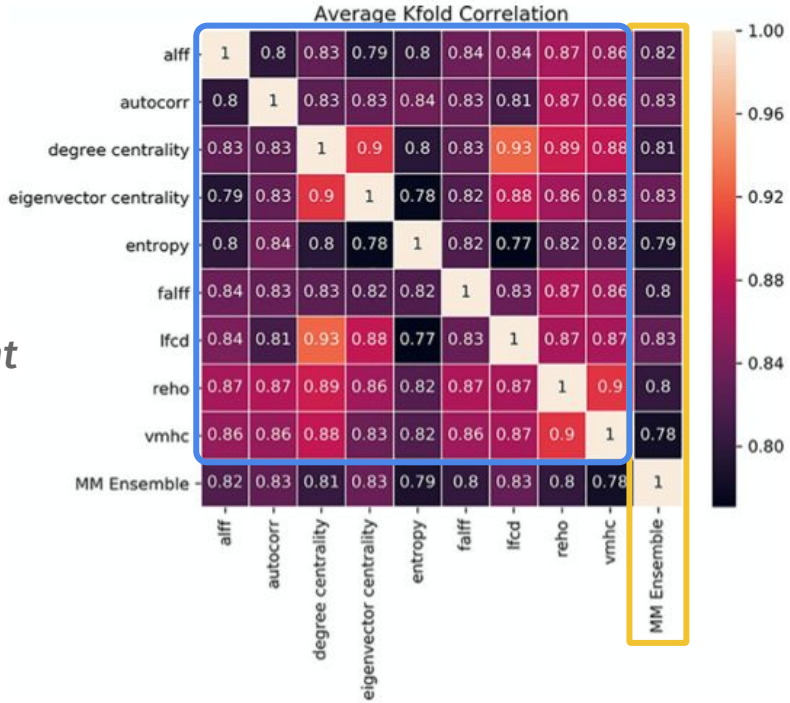
Leave-site-out CV

Kendall-correlations between models

Kendall-correlations between predictions of each SM-model & MM-ensemble

1) No cluster of high correlation between SM-models

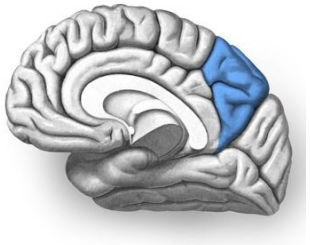
⇒ SM-models are processing independent information

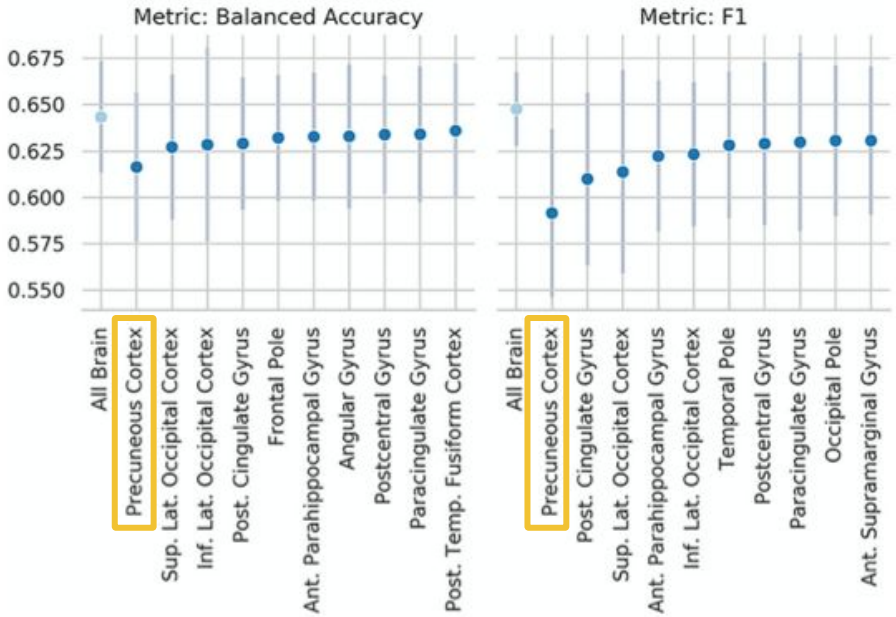


2) No highly correlated SM-model with MM-ensemble

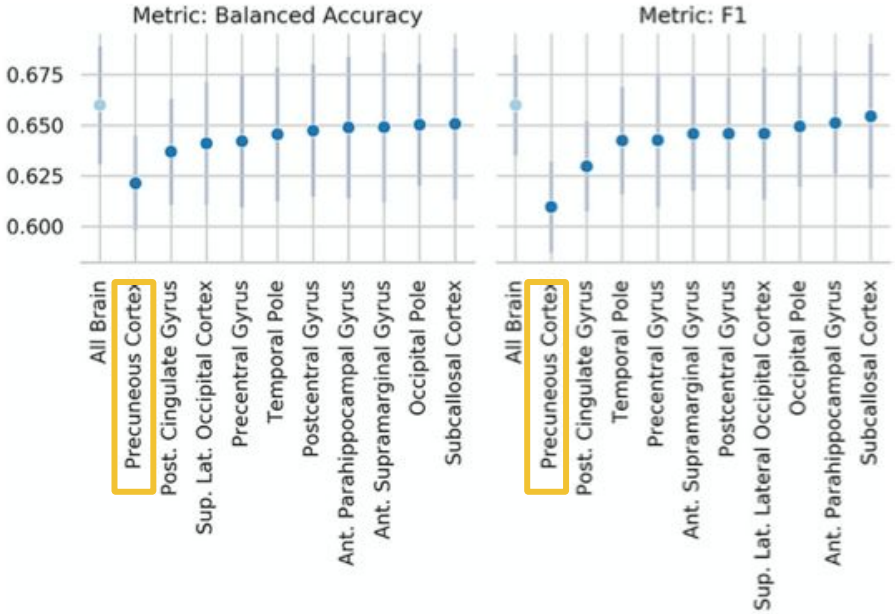
⇒ MM-ensemble are gathering information from each 9 SM-model

Occlusion Experiment





5-fold CV



Leave-site-out CV

Occlusion Experiment

frontiers in Neurology

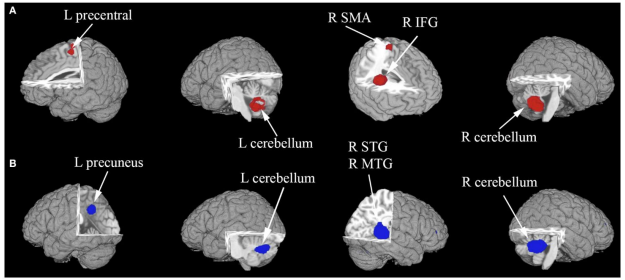
ORIGINAL RESEARCH
published: 24 July 2018
doi: 10.3389/fneur.2018.00556

Altered Resting-State Functional Activity in Patients With Autism Spectrum Disorder: A Quantitative Meta-Analysis

Wenhui Wang[1†], Jia Liu[2†], Shaojie Shi[1], Taiyuan Liu[1], Qiyong Gong[3] and Meiyun Wang[1]*

[1] Department of Radiology, Zhengzhou University People's Hospital and H
China, [2] Department of Radiology, Union Hospital of Tongji Medical Colleg
Wuhan, China, [3] School of Life Science and Technology, Xidian University,
Academy of Sciences, Beijing, China, [4] Huaxi MR Research Center, Depar
Sichuan University, Chengdu, China

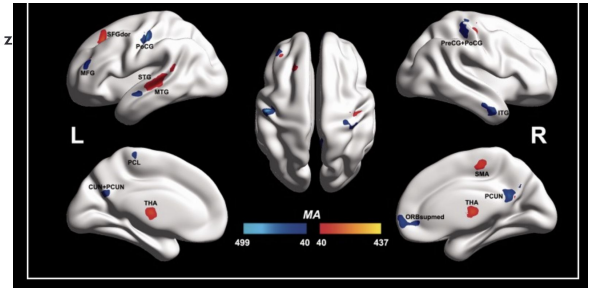
OPEN ACCESS



doi:10.1093/brain/awv051
BRAIN
A JOURNAL OF NEUROLOGY

Autism: reduced connectivity between cortical areas involved in face expression, theory of mind, and the sense of self

Wei Cheng,[1,*] Edmund T. Rolls,[2,3,*] Huaguang Gu,[4,*] Jie Z



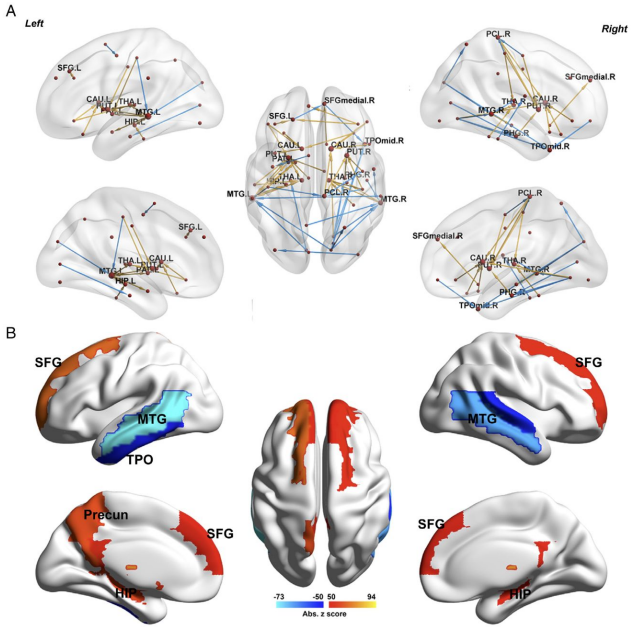
> Autism Res. 2020 Jan;13(1):32-44. doi: 10.1002/aur.2235. Epub 2019 Oct 27.

Effective connectivity in autism

Edmund T Rolls [1 2 3], Yunyi Zhou [3], Wei Cheng [3], Matthieu Gilson [4], Gustavo Deco [4 5], Jianfeng Feng [1 3]

Affiliations -

PMID: 31657

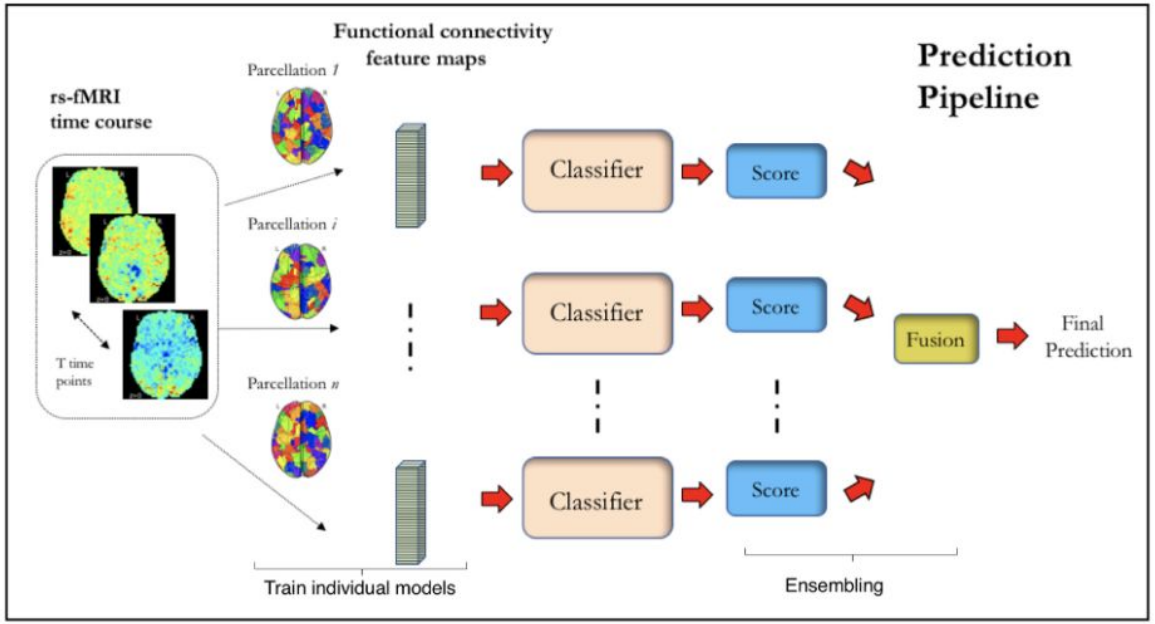


Discussion

More accurate prediction with an ensemble approach (MM-ensemble)

- ***MM-ensemble model*** takes advantage of multiple representation of the data, without being affected by increased noise.
- ***MM-model*** pooled together different informative summary measures, resulting in increased noise.
- While ***ReHo*** resulted in best accuracy, it requires selecting this measure *a priori*.

Possible explanations of low prediction accuracy



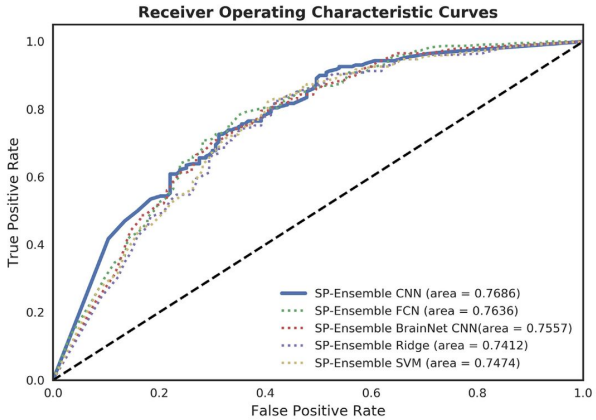


Fig. 4. ASD-HC Classification: Receiver Operating Curves for independent validation on ABIDE-2.

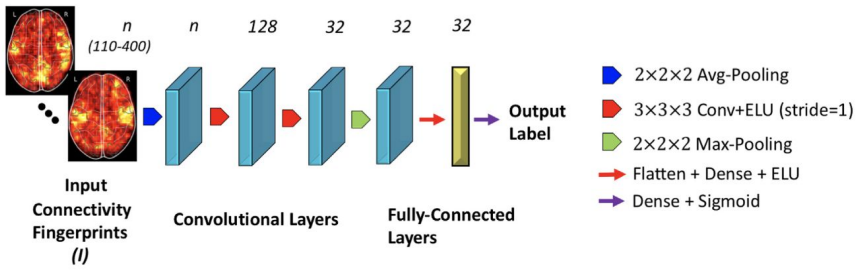
Khosla et al. (2019)

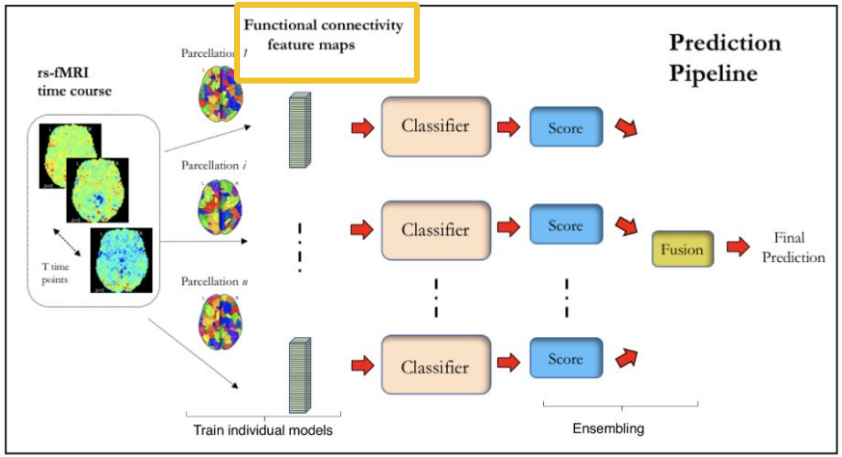
Possible explanations of low prediction accuracy

1. Heterogeneity of training dataset

2. Use of individual voxels rather than connectivity between ROIs

3. Reduction of convolution layers





Possible explanations of low prediction accuracy

1. Heterogeneity of training dataset

2. Use of individual voxels rather than connectivity between ROIs

3. Reduction of convolution layers

4. Model trained on different features

5. Difference in sample composition and data acquisition

6. Properties of time domain get lost when summarizing applied

Not much advantage of using 3D-CNN, instead of linear SVM

1. no low-dimensional patterns that capture the essence of ASD in summary measures

2. Insufficient amount of data for 3D-CNN

Thank you :)

References

Khosla M, Jamison K, Kuceyeski A, Sabuncu MR. Ensemble learning with 3D convolutional neural networks for functional connectome-based prediction. NeuroImage (2019) 199:651–62. doi: 10.1016/j.neuroimage.2019.06.012

Opelt A, Fussenegger M, Pinz A, Auer P. Weak Hypotheses and Boosting for Generic Object Detection and Recognition. Lecture Notes Comput Sci (2004) 3022:71–84. doi: 10.1007/978-3-540-24671-8_6

Wang Z, Li Y, Childress AR, Detre JA. Brain entropy mapping using fMRI. PloS One (2014) 9:e89948. doi: 10.1371/journal.pone.0089948

Woodward ND, Cascio CJ. Resting-State Functional Connectivity in Psychiatric Disorders. JAMA Psychiatry (2015) 72:743. doi: 10.1001/jamapsychiatry.2015.0484

Sherkatghanad Z, Akhondzadeh M, Salari S, Zomorodi-Moghadam M, Abdar M, Acharya UR, et al. Automated Detection of Autism Spectrum Disorder Using a Convolutional Neural Network. Front Neurosci (2020) 13:736. doi: 10.3389/fnins.2019.01325

Zang Y, Jiang T, Lu Y, He Y, Tian L. Regional homogeneity approach to fMRI data analysis. N eur oIma ge (2 00 4) 22 :394 –4 00 . doi: 10 .1 01 6/ j.neuroimage.2003.12.030

Zuo X-N, Di Martino A, Kelly C, Shehzad ZE, Gee DG, Klein DF, et al. The oscillating brain: complex and reliable. Neuroimage (2010) 49:1432–45. doi: 10.1016/j.neuroimage.2009.09.037

Zuo X-N, Ehmke R, Mennes M, Imperati D, Xavier Castellanos F, Sporns O, et al. Network Centrality in the Human Functional Connectome. Cereb Cortex (2012) 22:1862–75. doi: 10.1093/cercor/bhr269

Tomasi D, Volkow ND. Association between functional connectivity hubs and brain networks. Cereb Cortex (2011) 21:2003–13. doi: 10.1093/cercor/bhq268

Questions & Comments

Questions about 3D-CNN

Jeunghyun

It was interesting to read how 3D-CNN trained on a comparably moderate size of rsfMRI images performed pretty well. Couple questions following:

(1) Would it be possible to gain similar performance with healthy participants' fMRI dataset (e.g. ABCD)? I know not much about autism and its mechanism along with rsfMRI results, but I suppose it would be harder to apply the exact pipeline to compare fMRI results within a healthy group?

Jaeyeon

(1) In the discussion, the authors mention that "The amount of data available might not be sufficient to leverage the ability of CNNs to detect patterns. This could explain accuracies similar to those of much less flexible linear-SVM models." Does this mean that in practice, there wouldn't be a significant benefit of using CNN methods especially considering the cost? I think this is similar to what we discussed in previous journal clubs, but I wasn't sure if the authors were trying to imply this.

Cheol Jun

(2) Up to my knowledge, 3D-CNN is also affordable to employ XAI techniques, and we can employ XAI techs like LRP to find how each part of features is attributed to the inference. So what do you expect to find from this? What kinds of scenarios are possible to draw scientific implications by further interpreting the model behaviors?

Hyeonjin

(1) I found the reasons why they structured the inside of 3D-CNN as they applied are not compelling enough. When there should be almost infinite possible structures of models, is there a particular technique to choose a specific one?

Questions about prediction accuracy

Yoseph

(2) In the current study, the MM-ensemble seemed to perform well and was shown to be the overall best performing model. However, after applying the leave-site-out CV procedure, the performance dropped, suggesting its lack of generalizability. If such is the case, can we salvage the model by adding more (multisite) data?

Young

1. Not sure why authors did not use the raw data for generating prediction. Possibly due to computational costs? I wonder if using raw data will improve the prediction (I guess not much given its sample size and Schultz et al. results?).
2. Literature suggests that ASD is associated with altered self processing (e.g., altered neural activity while viewing one's own pictures) compared to typically developing participants. I wonder if using a resting-state fMRI with a picture viewing paradigm might increase the classification accuracy.

Questions about rs-fMRI summary measures

Jihyun

2. The authors summarized the temporal dimension with 9 measures. However, there has been an increase in the usage of temporal information in the resting-state data. Are there any approaches to reduce the dimension of spatial dimension? Would the classification accuracy increase if we can develop and apply an ensemble model with both temporal and spatially reduced data?

Cheol Jun

(1) In this study, the temporal dimension is reduced to 1 with specially designed features, e.g. ReHo, ALFF. Given the result that the accuracy is barely over 70%, do you think using raw temporal data would improve the performance? If so, what would be the bottleneck of implementing it? (The author is saying they are planning to try it on large datasets like UKBiobank.)

Youngha

1. I was curious about various functional connectivity measurements and summing over the time. Maybe they did it because those measurements are mainly informing how strongly the voxels are connected rather than how information is being communicated. In this regard, I think fALFF was the second efficient method to diagnose ASD. But this kind of power measuring would not be the novel method to find out features. How do you think of it? Would this kind of measurement and summing eventually be efficient after complementing?